Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2004
PI Name:	Gould, Harvey Allen Ph.D.		
Project Title:	Feasibility Study for an Electron Electric Dipole Moment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gould@lbl.gov	**Fax:**	FY 510-486-7981
PI Organization Type:	GOVERNMENT	**Phone:**	510-486-7777
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	MS 71-259		
PI Address 2:	LBNL		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	C73534 A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To determine if a slow atom microgravity experiment can improve the sensitivity to an electron electric dipole moment (EDM) by orders of magnitude. Develop and test experimental feacures such as electric field quantization, multiple quantum transitions, beam focusing and reflection, electric field ramping, pulsed oscillatory fields, state selection with Stark separated levels. Detemine sensitivity to systematic effects, sensitivity to electric and magnetic noiseetc. Will determine the most useful features to incorporate, identify problem areas, and help assess the feasibility of proceeding to flight definition studies for a microgravity electron electric dipole moment experiment.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Spinoff can improve accuracy of cesium fountain clocks
Task Progress:	Apparatus undergoing final assembly with all in-vacuum mechanical parts fabricated and cleaned and external parts in shops. Atom focusing results being written up for publication.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Articles in Peer-reviewed Journals	Amini JM, Gould H. "High precision measurement of the static dipole polarizability of cesium." Physical Review Letters 2003 Oct;91:153001. http://dx.doi.org/10.1103/PhysRevLett.91.153001 , Oct-2003
Presentation	Gould, H. "Search for CP-violation in the Lepton Sector: Electron Electric Dipole Moment (EDM) Experiments with Laser-Cooled Cesium Atoms" Spring Meeting of the American Physical Society May-2004
Presentation	Gould, H; Amini, J. "High-Precision Measurement of the Static Dipole Polarizability of Cs" Spring Meeting of the American Physical Society May-2004
Presentation	Gould, H; Amini,J. Kalnins, J "Electrostatic Focusing of Cesium Atoms in a Fountain" Annual Meeting of the APS Division of Atomic Molecular and Optical Physics May-2004